

ORD SHARP, NE, USA

WMO: 725524

Lat: 41.623N

Lon: 98.948W

Elev: 629

StdP: 93.99

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 94958

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-17.5	-24.0	0.5	-19.4	-21.7	0.6	-16.7	15.1	-1.6	13.8	-1.9	3.2	320	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	35.0	22.5	33.0	22.4	31.5	21.9	24.9	31.4	24.0	30.3	23.0	29.4	5.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	19.0	28.8	22.0	18.0	28.0	21.1	17.0	26.9	79.3	31.2	75.3	30.3	71.5	29.6	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.6	11.2	9.6	-25.8	37.5	3.0	2.1	-28.0	39.0	-29.7	40.3	-31.4	41.4	-33.5	42.9		
			-26.1	26.7	2.9	0.8	-28.1	27.3	-29.8	27.7	-31.5	28.1	-33.6	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	-3.8	-3.3	3.7	9.4	15.0	21.2	24.1	22.5	18.0	9.9	2.8	-3.5
	DBStd	11.32	6.46	6.65	6.31	5.35	4.97	3.71	3.23	3.16	4.59	5.17	5.71	6.08
	HDD10.0	1804	429	373	211	73	12	0	0	0	2	64	222	417
	HDD18.3	3634	687	606	456	271	127	16	2	4	61	263	466	676
	CDD10.0	1708	0	1	14	55	167	336	438	387	242	62	6	0
	CDD18.3	498	0	0	1	3	24	101	182	134	51	3	0	0
	CDH23.3	5603	0	1	17	72	342	1119	2026	1313	626	85	4	0
	CDH26.7	2305	0	0	2	17	112	425	956	541	229	23	0	0
Wind	WSAvg	4.0	4.1	4.3	4.6	5.1	4.5	3.9	3.2	3.1	3.6	3.9	4.0	4.0
Precipitation	PrecAvg	640	13	17	32	77	93	110	78	77	57	52	24	15
	PrecMax	903	38	36	76	222	202	268	198	172	122	142	75	75
	PrecMin	306	1	1	0	21	4	4	11	17	10	0	0	0
	PrecStd	115	9	9	24	44	51	59	50	43	30	35	20	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	18.1	25.5	29.1	32.6	35.2	37.9	36.7	33.9	30.0	23.2	14.9
		MCWB	7.4	8.4	13.0	16.1	18.8	21.7	22.6	22.1	20.5	17.4	12.1	6.9
	2%	DB	11.3	13.5	21.4	25.6	29.7	32.9	35.9	33.7	31.8	25.6	19.2	11.2
		MCWB	4.8	6.7	11.6	14.7	18.4	22.0	22.9	22.9	20.8	15.3	10.2	5.4
	5%	DB	8.0	9.8	17.9	22.4	27.3	31.2	33.7	31.7	29.1	22.5	15.8	8.1
		MCWB	3.3	4.6	10.3	13.4	17.5	21.4	23.2	22.7	19.8	13.8	8.8	3.5
	10%	DB	5.3	6.3	14.5	19.6	24.5	29.0	31.9	29.7	27.2	19.4	12.4	5.2
		MCWB	1.8	2.8	8.4	11.8	16.4	20.5	22.8	21.8	18.6	12.4	7.3	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	9.8	15.6	17.8	21.4	25.1	26.4	25.7	23.4	19.1	13.8	8.6
		MCDB	15.9	17.3	20.1	25.1	28.4	31.4	32.8	31.8	29.9	25.5	19.7	13.4
	2%	WB	5.2	6.8	13.3	15.9	20.0	23.6	25.2	24.6	22.0	16.8	11.4	5.6
		MCDB	10.6	13.2	19.1	23.2	26.9	30.3	31.9	30.7	29.0	22.6	17.1	10.8
	5%	WB	3.7	4.9	10.7	14.2	18.7	22.5	24.3	23.5	21.0	15.1	9.4	3.8
		MCDB	7.7	9.2	17.4	21.0	25.7	29.2	30.8	29.5	27.3	20.4	15.5	7.6
	10%	WB	2.0	2.9	8.4	12.6	17.2	21.5	23.4	22.5	19.8	13.4	7.3	1.9
		MCDB	5.0	6.2	14.7	18.9	23.5	27.8	29.9	28.2	25.3	18.9	12.1	4.8
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.5	13.8	13.6	13.1	12.8	13.1	12.8	14.5	14.4	14.1	12.1
		MCDBR	17.0	17.9	20.5	19.1	16.8	15.1	15.9	15.1	16.6	19.9	20.2	16.7
		MCWBR	11.3	11.6	11.7	10.1	8.0	6.4	6.3	6.6	7.5	10.6	12.0	11.4
	5% WB	MCDBR	16.0	17.0	18.7	16.3	14.6	12.9	13.5	13.0	14.1	15.7	18.7	16.1
		MCWBR	11.0	11.3	11.2	9.0	7.7	6.6	6.5	6.4	7.0	9.2	11.5	11.5
Clear-Sky Solar Irradiance	taub	0.272	0.278	0.303	0.340	0.365	0.380	0.397	0.395	0.358	0.319	0.277	0.268	
	taud	2.478	2.485	2.454	2.366	2.382	2.375	2.360	2.354	2.410	2.499	2.530	2.490	
	Ebn at Noon	893	942	950	929	910	892	873	865	876	879	882	868	
	Edh at Noon	73	85	100	117	119	120	120	118	104	84	69	67	
All-Sky Solar Radiation	RadAvg	2.14	3.05	4.04	5.10	5.94	6.83	6.80	5.83	4.73	3.34	2.35	1.83	
	RadStd	0.18	0.17	0.36	0.27	0.42	0.46	0.26	0.34	0.30	0.35	0.17	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page